ps1-fx3-cart. A PS1 expansion-port (PIO) cartridge: an FX3 bridges the PIO bus to USB 3 through GPIF II,
holds the console in reset until its firmware runs, and can carry an optional BIOS NOR flash. Design notes: docs/design.md.

PIO

File: pio.kicad_sch

Flash

File: flash.kicad_sch

FX3

File: fx3.kicad_sch

USB

File: usb.kicad_sch

Power

File: power.kicad_sch

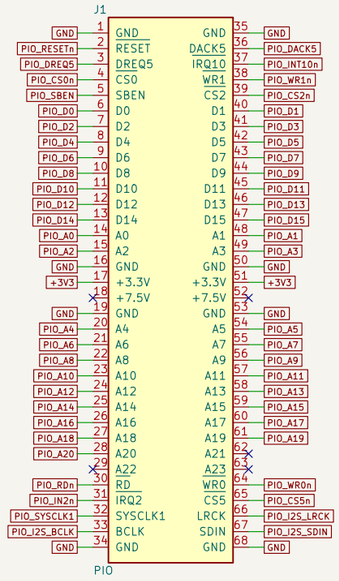
rixnobis

Sheet: /
File: ps1-fx3-cart.kicad_sch

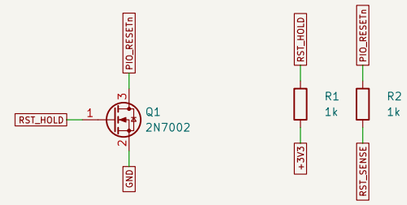
Title: ps1-fx3-cart: PS1 PIO to USB 3

Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.7		Id: 1/6

PIO expansion port (CN103), pin numbers as psx–spx (console rear view). On a stock console pin 5 (SBEN) and pin 39 (/CS2) are GND and pins 31/65 are NC; they carry SBEN, /CS2, /IRQ2 and /CS5 only with a Switch Board mod. The 3.3 V pins feed the whole cart; 7.5 V and A21–A23 are unused.



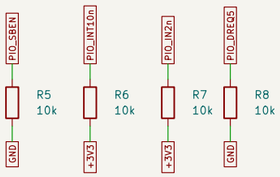
Reset hold: Q1 pulls PIO /RESET low while its gate is high. The gate pin GPIO33 is tristated by the boot ROM (AN76405 Table 30), so the 1k pull–up holds the console in reset until FX3 firmware drives GPIO33 low. Sizing: the FX3 can put a 10k pull–down on any pin (datasheet I/O states), and AN76405 does not say the boot ROM's tristate excludes it; 1k against 10k still gives 3.3 V x 10/11 = 3.0 V on the gate, above the 2N7002's 2.5 V maximum VGS(th). Released, GPIO33 sinks 3.3 mA, under the IOL of quarter drive (3.96 mA even at VIO 2.5 V, datasheet Table 9). GPIO34 senses /RESET through 1k: 3.3 mA at most if it is ever driven as an output, and an internal 10k pull–down still reads high (3.0 V).



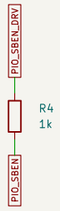
/CS2 sense: GPIO39 reads pin 39 through 1k. Pin 39 is GND on a stock console, so GPIO39 misdriven high sinks 3.3 mA instead of shorting the pin.



Defaults while the FX3 is unconfigured: SBEN low (no Switch Board takeover), /IRQ10 and /IRQ2 high (inactive), DREQ5 low (inactive; DREQ/DACK are active high).



SBEN drive: GPIO37 reaches pin 5 through 1k. Pin 5 is GND on a stock console, so GPIO37 driven high sinks 3.3 mA instead of shorting the pin.



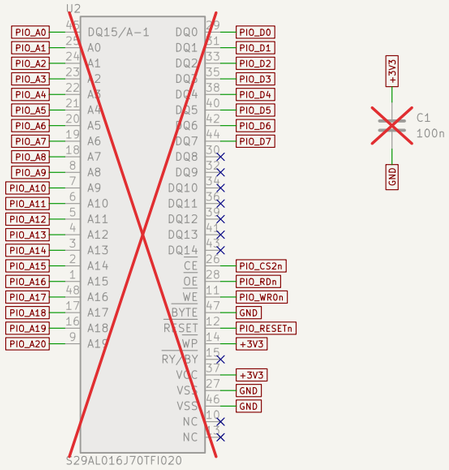
rixnobis

Sheet: /PIO/
File: pio.kicad_sch

Title: PIO connector, reset hold, defaults

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/6

Optional BIOS NOR flash, NOT FITTED by default. S29AL016J (2 MB) in byte mode: BYTE# low, PIO A0 on DQ15/A−1, PIO A1−A20 on flash A0−A19, so flash byte address = PIO byte address and the unlock cycles land at BIOS offsets 0xAAA/0x555. /CE = /CS2, /OE = /RD, /WE = /WR0, D0−D7 shared with the FX3. No flash signal goes to the FX3 (GPIO39 only senses /CS2). The full 2 MB needs the BIOS window (1F801010) enlarged to 21 address bits. Fit only with the Switch Board /CS2 mod: on a stock console pin 39 is GND, which would enable the flash on every /RD. RESET# follows the console reset; WP# high; RY/BY# unused (firmware polls DQ7/DQ6).



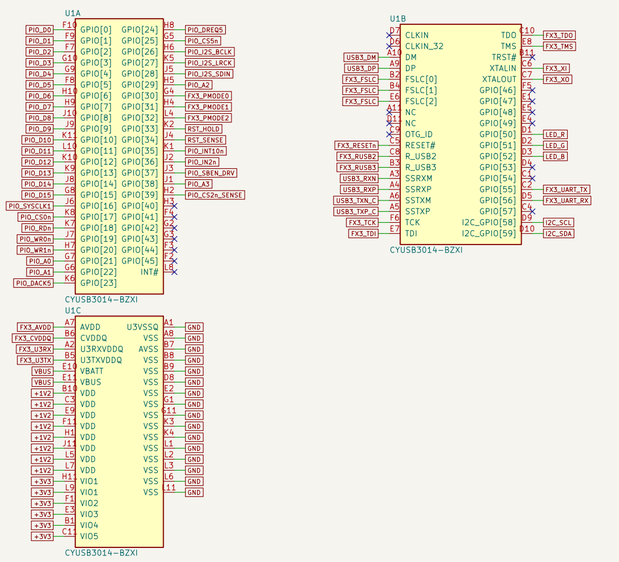
rixnobis

Sheet: /Flash/
File: flash.kicad_sch

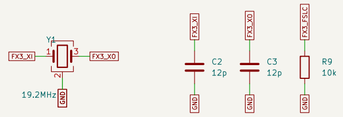
Title: Optional BIOS NOR flash (not fitted)

Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.7		Id: 3/6

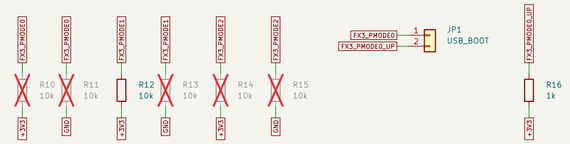
CYUSB3014. GPIF II in 16-bit mode takes the PIO bus: DQ[15:0]=D15..D0, PCLK=SYSCLK1, strobes, selects, A0-A2, DREQ5/DACK5 and I2S on CTL[0..12] (VIO1). Slow lines on VIO2 GPIO33-39. All VIO on the PIO 3.3 V rail. Map: PIO_GPIO in tools/gen_sch.py, docs/design.md.



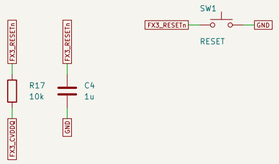
19.2 MHz crystal, FSLC[2:0]=000 through one 10k (AN70707 Figure 9). No series resistor on the crystal (AN70707). Load caps sized for a 12 pF crystal with ~5 pF stray.



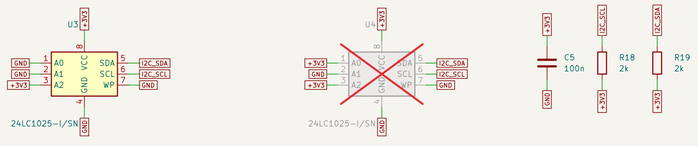
Boot: PMODE F1F = I2C EEPROM, USB fallback (AN70707 Table 9). 10k up/down footprints on every PMODE pin, only PMODE1 up fitted. Jumper JP1 pulls PMODE0 high = F11, USB boot only.



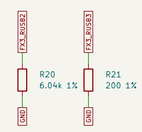
Reset: RC holds RESET# low > 1 ms after power (tRPW 1 ms, tRR 5 ms with a crystal); button and the JTAG header can also pull it.



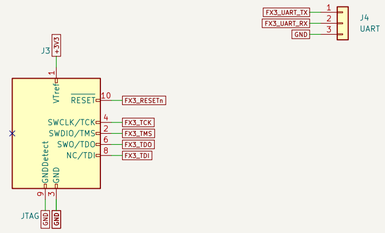
I2C: boot EEPROM, 2k pull-ups to VIO5 (AN70707). 24LC1025 A2 must be tied high; A1:A0 = 00.



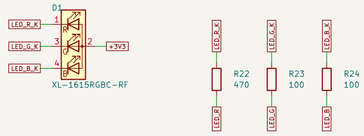
USB reference resistors (AN70707): R_USB2 6.04k 1%, R_USB3 200R 1%.



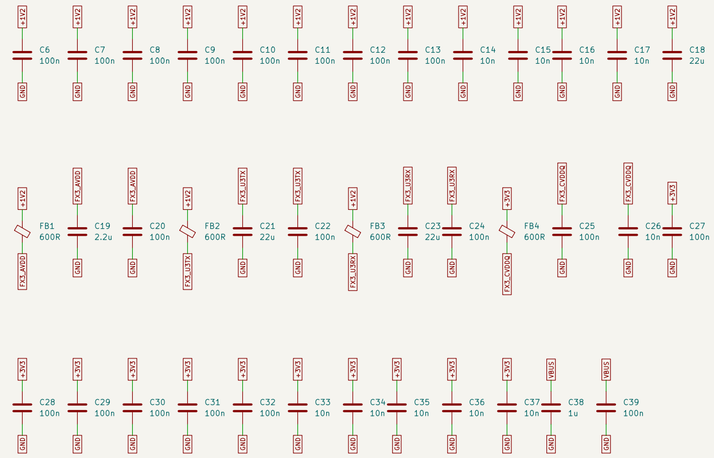
Debug: JTAG (no external pulls; FX3 has internal ones). UART on GPIO55/56.



Status LED: common-anode RGB on +3V3, cathodes sunk by GPIO50/51/52 (the unused I2S balls). Pin order per the XL-1615RGBC-RF datasheet: 1 R, 2 common anode, 3 G, 4 B. States in docs/design.md.

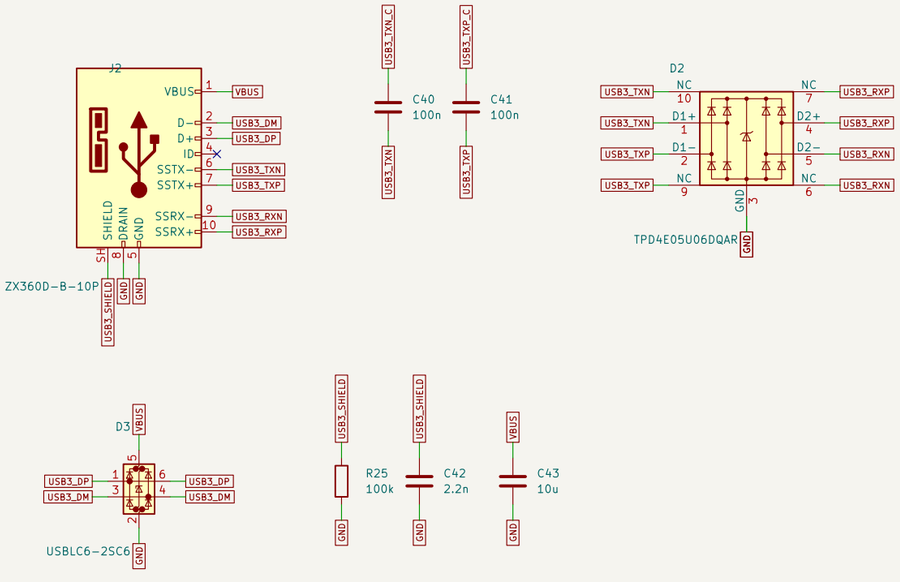


Decoupling per AN70707 Table 3. AVDD, U3TXVDDQ, U3RXVDDQ and CVDDQ each behind a ferrite (AN70707 Figure 1, Appendix B): 22 uF on each USB 3 supply (datasheet).



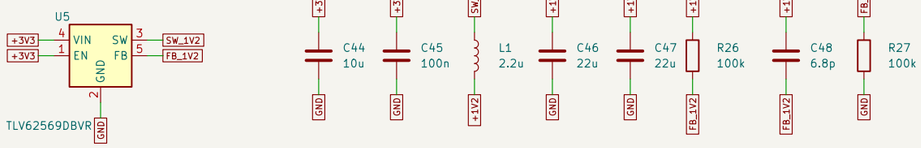
rixnobis	
Sheet: /FX3/	
File: fx3.kicad_sch	
Title: FX3 USB 3 bridge	
Size: A2	Date:
KiCad E.D.A. 10.0.7	Rev: A Id: 4/6

USB 3.0 Micro-B, Hirose ZX360D-B-10P. PROVISIONAL: connector choice open (docs/design.md); swap usb_micro_b() in tools/gen_sch.py. SS TX AC-coupled with 100 nF, ESD on the receptacle side of the caps. Pairs wired P to P; AN70707 p.22 allows a P/N swap on SS RX if the layout wants it. ID open (device). Shell to GND through 100k || 2.2 nF (AN70707 Figure 35). VBUS only reaches the FX3's VBUS detect.



rixnobis		
Sheet: /USB/		
File: usb.kicad_sch		
Title: USB 3 connector (provisional: Micro-B)		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 5/6

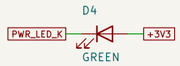
Supply: the PIO 3.3 V rail (pins 17/51, 3.3 V nominal, up to 3.5 V) feeds every FX3 I/O domain directly (VIO1–5 and CVDDQ, 1.7–3.6 V operating). TLV62569 buck makes the 1.2 V core and USB 3 supplies from it (SLVSDG0: VFB 0.6 V, R1 = R2 = 100k, 6.8 pF feed-forward for R2 = 100k, 2.2 uH, 4.7 uF in minimum, 10–47 uF out). EN tied high: the core runs whenever the console is on.

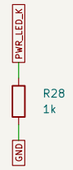


Power flags (ERC): +1V2, +3V3, FX3_AVDD, FX3_CVDDQ, FX3_U3RX, FX3_U3TX



Power LED.





rixnobis

Sheet: /Power/
File: power.kicad_sch

Title: 1.2 V buck from the PIO 3.3 V rail

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 6/6